Amazon Sales Performance Analysis Report

1. Project Overview

This report analyzes Amazon sales performance, product category revenue, state-wise sales distribution, monthly sales trends, product performance, order status, and fulfillment patterns. The analysis is based on the Amazon Sale Report dataset, processed using Python for data cleaning and exploratory data analysis, PostgreSQL for SQL-based business analysis, and Power BI for dashboard visualization.

The main objective is to identify high-performing product categories, top-revenue states, best-selling products, and sales trends to support data-driven business decisions.

2. Dataset Summary

- **Dataset Name:** Amazon Sale Report.csv
- **Initial Rows:** 128,975
- **Initial Columns:** 24
- **Columns After Cleaning:** 23
- **Key Features:**
    - **Transaction Details:** Order ID, Date, Status
    - **Order Fulfillment:** Fulfilment, Sales Channel, ship-service-level, fulfilled-by
    - **Product Information:** SKU, ASIN, Style, Category, Size
    - **Sales Information:** Qty, Amount, currency
    - **Shipping Details:** ship-city, ship-state, ship-postal-code, ship-country
    - **Additional Information:** Courier Status, promotion-ids, B2B
- **Duplicate Records:** No duplicate rows identified.
- **Missing Data:** Missing values were identified in Amount, shipping details, promotion information, fulfillment information, and other columns.

3. Exploratory Data Analysis Using Python

The dataset was processed and analyzed using Python and pandas.

- **Data Loading:** Imported the Amazon sales dataset using pandas.

- **Initial Exploration:** Used df.shape, df.head(), df.info(), and df.describe() to understand the dataset structure and statistical summary.

```
Dataset size: (128975, 24)
Duplicate rows: 0
```

[26]:

	index	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	...	currency	Amount	ship-city	ship-state	ship-postal-code	c
0	0	405-8078784-5731545	04-30-22	Cancelled	Merchant	Amazon.in	Standard	SET389	SET389-KR-NP-S	Set	...	INR	647.62	MUMBAI	MAHARASHTRA	400081.0	
1	1	171-9198151-1101146	04-30-22	Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard	JNE3781	JNE3781-KR-XXXL	kurta	...	INR	406.00	BENGALURU	KARNATAKA	560085.0	
2	2	404-0687676-7273146	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3371	JNE3371-KR-XL	kurta	...	INR	329.00	NAVI MUMBAI	MAHARASHTRA	410210.0	
3	3	403-9615377-8133951	04-30-22	Cancelled	Merchant	Amazon.in	Standard	J0341	J0341-DR-L	Western Dress	...	INR	753.33	PUDUCHERRY	PUDUCHERRY	605008.0	
4	4	407-1069790-7240320	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3671	JNE3671-TU-XXXL	Top	...	INR	574.00	CHENNAI	TAMIL NADU	600073.0	

5 rows × 24 columns

- **Data Cleaning:** Removed the unnecessary Unnamed: 22 column, reducing the dataset from 24 to 23 columns.
- **Missing Values:** Filled missing shipping city, state, country, and fulfillment values with "Unknown", and missing promotion IDs with "No Promotion".

```
Dataset shape: (128975, 23)
Duplicate rows: 0
Missing values:
index                    0
Order ID                 0
Date                     0
Status                   0
Fulfilment               0
Sales Channel            0
ship-service-level       0
Style                    0
SKU                      0
Category                 0
Size                     0
ASIN                     0
Courier Status           0
Qty                      0
currency              7795
Amount                7795
ship-city                0
ship-state               0
ship-postal-code         0
ship-country             0
promotion-ids            0
B2B                      0
fulfilled-by             0
dtype: int64
```

- **Date Processing:** Converted the Date column into datetime format and checked invalid dates.
- **Numeric Conversion:** Converted Amount and Qty columns into numeric data types.
- **Missing Amount Analysis:** Investigated records with missing Amount values using order status, quantity, and currency.

- **Courier Status Cleaning:** Replaced missing Courier Status values with "Unknown".
- **Postal Code Cleaning:** Converted postal codes to string values and replaced missing entries with "Unknown".
- **Data Quality Validation:** Checked dataset dimensions, duplicate records, missing values, and data types.
- **Sales Analysis:** Calculated total recorded sales, average recorded sales amount, and the number of unique orders with recorded amounts.
- **Category Analysis:** Aggregated total sales, average sales, and unique order counts by product category.
- **State Analysis:** Calculated total sales and order counts for each shipping state.
- **Monthly Sales Analysis:** Grouped sales by month to identify changes in revenue and order counts.
- **Cleaned Dataset:** Exported the processed dataset as Amazon_Sales_Cleaned.xlsx.

4. Data Analysis Using SQL (Business Transactions)

The cleaned dataset was imported into PostgreSQL to perform structured business analysis.

- **Database Name:** Amazon_Sales_db
- **Table Name:** sale
- **Database Integration:** Connected Python to PostgreSQL using SQLAlchemy and psycopg2.

The following business questions were addressed using SQL queries:

1. **Total Sales and Quantity:** Calculated total recorded sales, average recorded sales amount, and total quantity sold.

	total_sales numeric	average_sales numeric	total_quantity_sold numeric
1	78592678.30	648.56	116482

2. **Category Performance:** Identified product categories generating the highest sales revenue.

	Category text	total_sales numeric
1	Set	39204124.03
2	kurta	21299546.70
3	Western Dre...	11216072.69
4	Top	5347792.30
5	Ethnic Dress	791217.66
6	Blouse	458408.18
7	Bottom	150667.98
8	Saree	123933.76
9	Dupatta	915.00

Total rows: 9 Query complete 00:

3. **State-Wise Sales:** Identified the top 10 states by total recorded sales.

	ship-state text	total_sales numeric
1	MAHARASHTRA	13335534.14
2	KARNATAKA	10481114.37
3	TELANGANA	6916615.65
4	UTTAR PRADESH	6816642.08
5	TAMIL NADU	6515650.11
6	DELHI	4235215.97
7	KERALA	3830227.58
8	WEST BENGAL	3507880.44
9	ANDHRA PRADE...	3219831.72

Total rows: 10 Query complete 00:0

4. **Monthly Sales Trend:** Analyzed monthly changes in sales revenue.

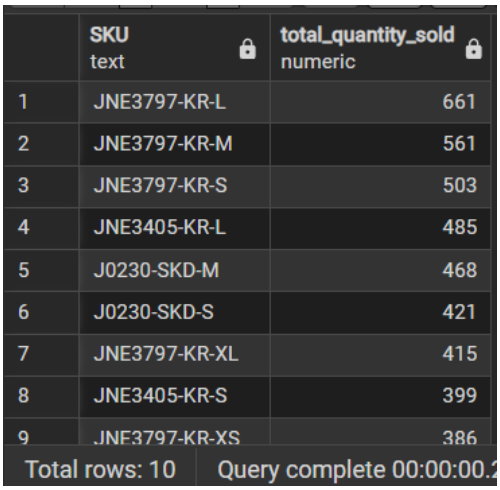
	month timestamp without time zone	total_sales numeric
1	2022-03-01 00:00:00	101683.85
2	2022-04-01 00:00:00	28838708.32
3	2022-05-01 00:00:00	26226476.75
4	2022-06-01 00:00:00	23425809.38

5. **Top Revenue-Generating Products:** Identified the top 10 SKUs by total recorded sales.

	SKU text	total_sales numeric
1	J0230-SKD-M	527699.20
2	JNE3797-KR-L	524581.77
3	J0230-SKD-S	479937.14
4	JNE3797-KR-M	454290.16
5	JNE3797-KR-S	407302.57
6	JNE3797-KR-XL	332155.24
7	J0230-SKD-L	305616.95
8	JNE3797-KR-XS	303616.70
9	SET268-KR-NP-XL	284058.96

Total rows: 10 Query complete 00:0

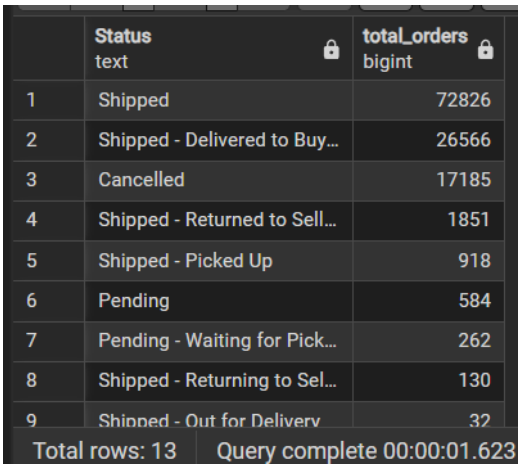
6. **Quantity by Product:** Identified the products with the highest quantities sold.



	SKU text	total_quantity_sold numeric
1	JNE3797-KR-L	661
2	JNE3797-KR-M	561
3	JNE3797-KR-S	503
4	JNE3405-KR-L	485
5	J0230-SKD-M	468
6	J0230-SKD-S	421
7	JNE3797-KR-XL	415
8	JNE3405-KR-S	399
9	JNE3797-KR-XS	386

Total rows: 10 Query complete 00:00:00.2

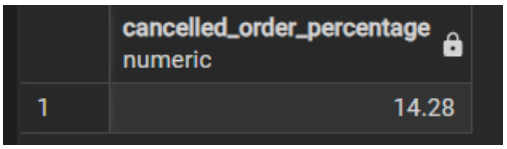
7. **Order Status Analysis:** Counted unique orders by status.



	Status text	total_orders bigint
1	Shipped	72826
2	Shipped - Delivered to Buy...	26566
3	Cancelled	17185
4	Shipped - Returned to Sell...	1851
5	Shipped - Picked Up	918
6	Pending	584
7	Pending - Waiting for Pick...	262
8	Shipped - Returning to Sel...	130
9	Shipped - Out for Delivery	32

Total rows: 13 Query complete 00:00:01.623

8. **Cancelled Order Percentage:** Calculated the percentage of unique orders with a cancelled status.



	cancelled_order_percentage numeric
1	14.28

9. **Average Sales by State:** Identified states with average recorded sales amounts greater than ₹1,000.



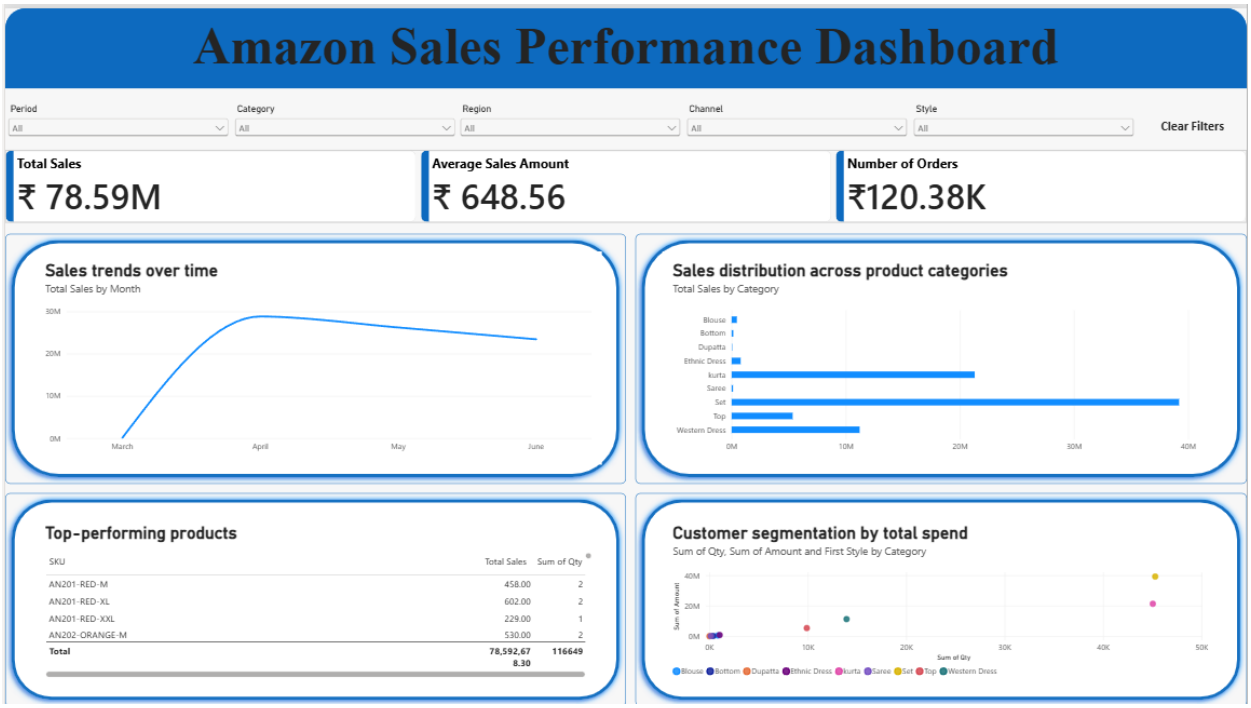
	ship-state text	average_sales numeric
1	bihar	1432.00

10. **Fulfillment Analysis:** Compared total recorded sales across fulfillment methods.

	Fulfilment text	total_sales numeric
1	Amazon	54322151.00
2	Merchant	24270527.30

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Focus on Top Categories:**
  - Prioritize product categories that generate the highest recorded sales.
  - Review category-level order counts and average sales to identify opportunities for improvement.
- **Expand High-Performing States:**
  - Study the top 10 states by sales to understand where demand is strongest.
  - Improve product availability and targeted marketing in promising markets.
- **Optimize Product Performance:**

    - Monitor the top revenue-generating SKUs and products with high quantities sold.
    - Use sales and quantity trends to support inventory planning.
- **Improve Order Management:**
    - Monitor cancelled orders and investigate common cancellation patterns.
    - Review order status and fulfillment data to identify potential operational issues.
- **Improve Data Quality:**
    - Investigate missing Amount values before preparing complete revenue reports.
    - Validate dates, quantities, and other important fields during future data processing.
- **Optimize Promotions:**
    - Analyze promotion IDs alongside sales performance to evaluate promotional activity.
    - Compare product and category sales during promotional periods where sufficient data is available.
- **Plan Inventory Using Monthly Trends:**
    - Use monthly sales and quantity patterns to support stock planning.
    - Review high-demand and low-demand periods before making inventory or marketing decisions.

7. Tools and Technologies Used

- **Python:** Data cleaning and exploratory data analysis.
- **Pandas:** Data manipulation, aggregation, and missing-value handling.
- **Excel:** Storage and review of the cleaned dataset.
- **PostgreSQL:** Database management and SQL business analysis.
- **SQLAlchemy and psycopg2:** Database connection and data transfer.
- **Power BI:** Interactive dashboard development and visualization.

8. Project Conclusion

The Amazon Sales Performance Analysis project demonstrates an end-to-end data analytics workflow, including data cleaning, exploratory analysis, database integration, SQL querying, and dashboard reporting.

Python was used to clean and analyze the dataset, PostgreSQL was used to answer business questions, and Power BI provides a visual way to communicate the results. The analysis supports better understanding of product category performance, state-wise sales, monthly trends, order status, and fulfillment patterns.

The project also highlights the importance of handling missing sales amounts carefully to ensure reliable reporting and informed business decisions.